Electromagnetic Power Density

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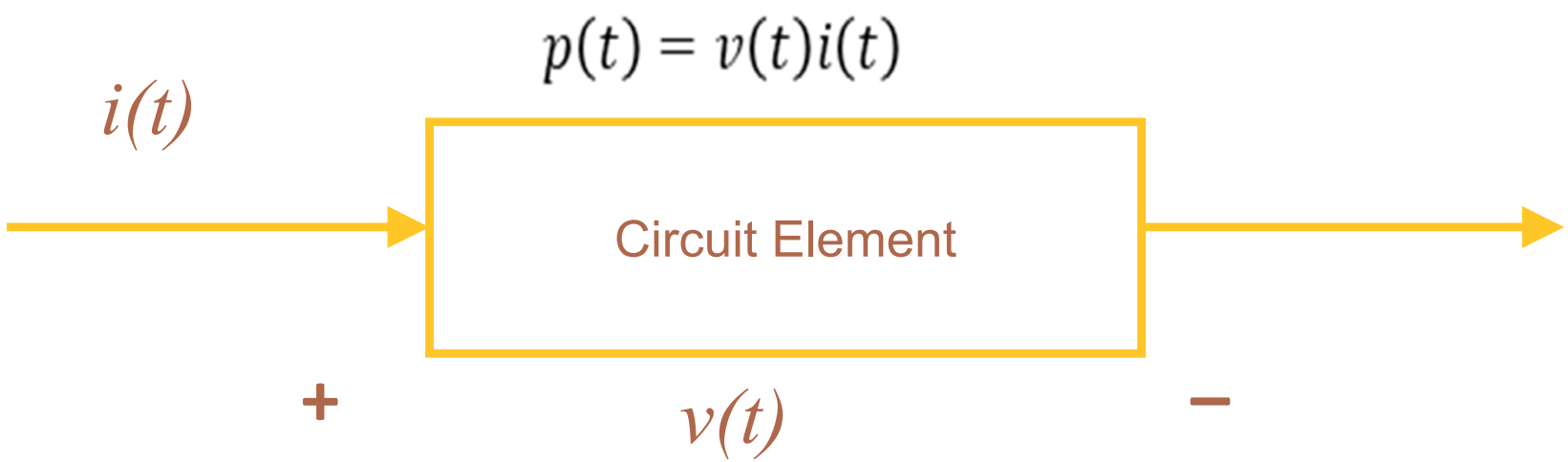
Lecture Objective

By the end of this lecture, you will be prepared to:

- **Analyze time-average power flow for electromagnetic plane waves.**

Power Concepts from Circuits

- Instantaneous power absorbed by a circuit element, $p(t)$:



Power Concepts from Circuits Continued

- Average power absorbed by a circuit element, P:

$$\boxed{P_{av} = \lim_{T \to \infty} \frac{1}{T} \int_0^T v(t) \cdot i(t) dt}$$

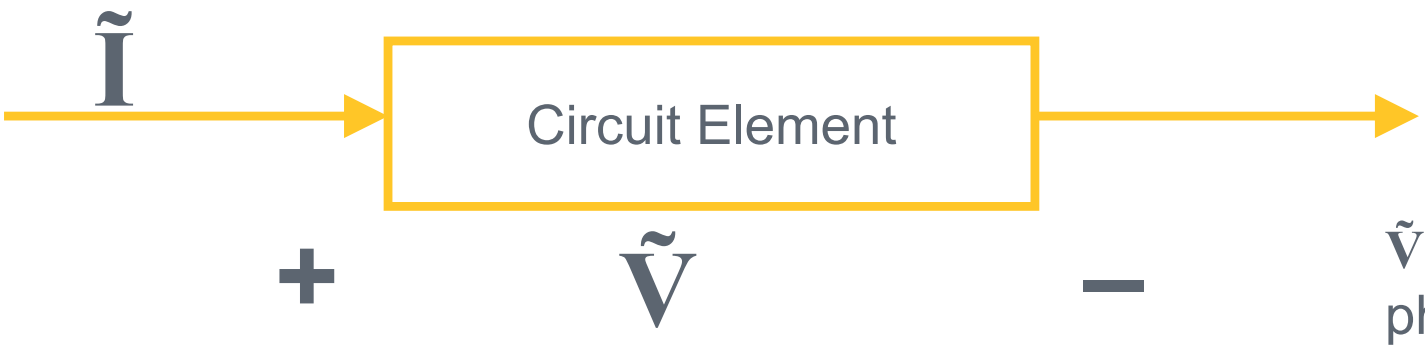
Question 1

What is the physical meaning of the quantity

$$\int_0^T v(t)\, i(t)\, dt$$

Power Concepts from Circuits Continued

- In a linear circuit, excited by sinusoidal source(s), the average power absorbed by a circuit element, P:



$\tilde{\mathbf{V}}$ and $\tilde{\mathbf{I}}$ are phasor voltage and current respectively.

$$P_{av} = \frac{1}{2}\mathrm{Re}\left\{\tilde{\mathbf{V}}\,\tilde{\mathbf{I}}^*\right\} = \frac{1}{2}\mathbf{V}\,\mathbf{I}\cos\left(\theta_v - \theta_i\right) = \frac{1}{2}\mathbf{V}\,\mathbf{I}\cos\left(\theta_z\right)$$

Poynting Vector

- Define the Poynting vector for electromagnetic wave as:

$$\vec{S}(\vec{R},t) = \vec{E}(\vec{R},t) \times \vec{H}(\vec{R},t)$$

- Notice that:
  - The Poynting vector has units of power density ($\mathrm{W/m^2}$)
  - The Poynting vector points in the direction of wave propagation.

Poynting's Theorem

- Poynting's theorem is a statement of conservation of energy:

$$-\oint_S \tilde{\vec{S}} \bullet d\vec{s} = \frac{\partial W_m}{\partial t} + \frac{\partial W_e}{\partial t} + P_d$$

- where:

$$-\oint_S \tilde{\vec{S}} \bullet d\vec{s} = \text{ net power entering the region}$$

$$W_m = \text{ total magnetic energy stored in the region}$$

$$W_e = \text{ total electric energy stored in the region}$$

$$P_d = \text{ total power dissipated in the region}$$

Poynting's Theorem Continued

- The net power entering a closed region is equal to the sum of the time rate of change of the magnetic energy stored in the region, the time rate of change of the electric energy in the region, and the power dissipated in the region.

Time-Average Poynting Vector for Time-Harmonic Waves

- The time-average Poynting vector for a time-harmonic wave is given by

$$\vec{S}_{av} = \frac{1}{T}\int_0^T \vec{S}\,dt = \frac{1}{2}\mathrm{Re}\left\{\tilde{\vec{E}} \times \tilde{\vec{H}}^*\right\}$$

Phasor representations of electric and magnetic field.

Time-Average Poynting Vector for Time-Harmonic Waves Continued

- Assume that the phasor representation of the electric field is given by

$$\tilde{\vec{E}} = \hat{x} E_0 e^{-\alpha z} e^{-j\beta z}$$

- Corresponding magnetic field is given by

$$\tilde{\vec{H}} = \frac{1}{\eta} \hat{z} \times \tilde{\vec{E}} = \hat{y} \frac{E_0}{\eta} e^{-\alpha z} e^{-j\beta z}$$

Time-Average Poynting Vector for Time-Harmonic Waves Continued

- We have

$$\eta = |\eta| e^{j\theta_\eta} \Rightarrow \eta^* = |\eta| e^{-j\theta_\eta}$$

$$\tilde{\vec{E}} \times \tilde{\vec{H}}^* = \hat{z} \frac{|E_0|^2}{\eta^*} e^{-2\alpha z} = \hat{z} \frac{|E_0|^2}{|\eta|} e^{j\theta_\eta} e^{-2\alpha z}$$

- So

$$\vec{S}_{av} = \hat{z} \frac{|E_0|^2}{2|\eta|} e^{-2\alpha z} \cos\theta_\eta$$

Question 2

- **Specialize the equation for time-average Poynting vector for time-harmonic plane wave to the lossless case**

Decibel Scale for Wave Attenuation

- The logarithmic decibel scale is often used in engineering to express power ratios.
- Consider the ratio of power density in an electromagnetic wave at some position (z) relative to a reference position ($z=0$):

$$\frac{S_{av}(z)}{S_{av}(0)} = e^{-2\alpha z}$$

Decibel Scale for Wave Attenuation Continued

- In decibels we have

$$10\log_{10}\left(\frac{S_{av}(z)}{S_{av}(0)}\right) = 10\log_{10}(e^{-2\alpha z})$$

$$= -20\alpha z\log_{10}(e^{-1})$$

$$= -(8.686\alpha)z$$

Summary

- Circuit theory:

$$p(t) = v(t)i(t)$$

$$P_{av} = \frac{1}{2}\mathrm{Re}\left\{\tilde{V}\,\tilde{I}^*\right\}$$
$$= \frac{|V|^2}{2|Z|}\cos\theta_z$$

- EM Field Theory:

$$\vec{S}(\vec{R},t) = \vec{E}(\vec{R},t) \times \vec{H}(\vec{R},t)$$

$$\tilde{\vec{S}}_{av} = \frac{1}{2}\mathrm{Re}\left\{\tilde{\vec{E}} \times \tilde{\vec{H}}^*\right\}$$
$$= \hat{z}\frac{|E_0|^2}{2|\eta|}e^{-2\alpha z}\cos\theta_\eta$$

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